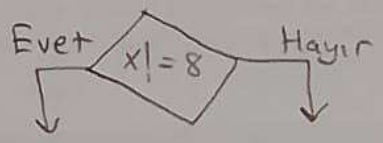
— Algoritma Ve Programlama —

Bir algoritma sıralı adımlar bütünüdür

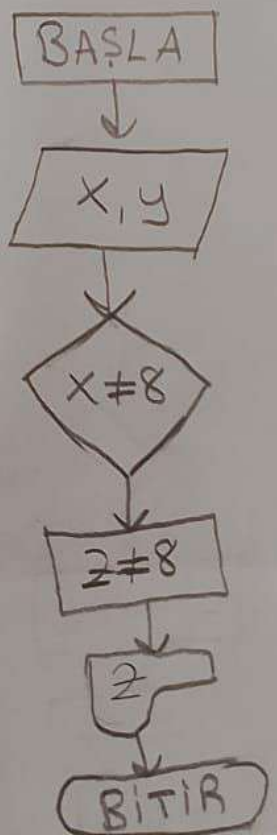
1. Sayı 1'i al 5 al
2. Sayı 2'yi al 0 al
3. Sayı 2 0'a eşitse 2. adıma git
4. Sayı 3 = Sayı 1 / Sayı 2
5. Sayı 3'ü yazdır.

Düzyazı Gösterim

1- BAŞLA
2- Birinci sayıyı gir.
3- İkinci sayıyı gir.
4- İki sayıyı toplayıp üçüncü sayıya ata.
5- Üçüncü sayı
6- BİTİR



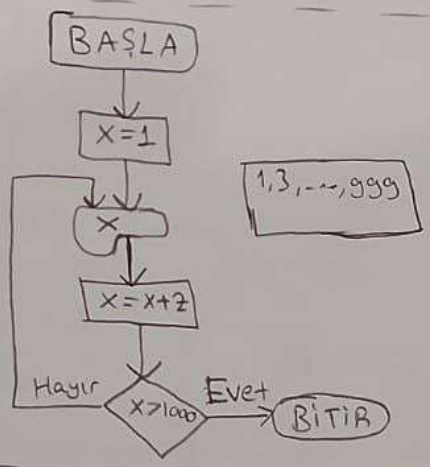
< > <= >= == !=

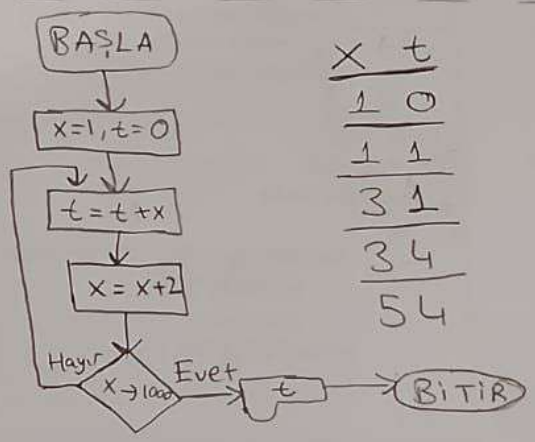


NOT: C programlama dilinde ^ (şapka) işlemi ÜS almada kullanılmaz!

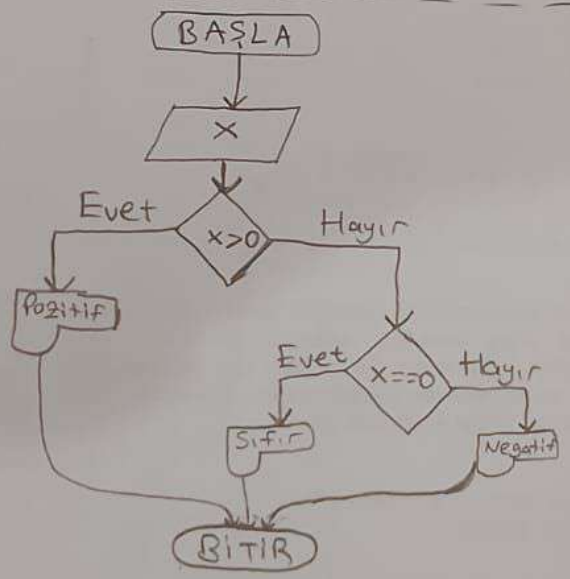
assignment = atama

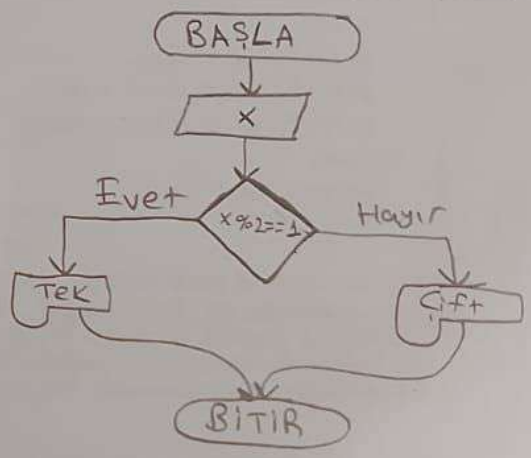
```
int x=5;      int z=x;
int y=8;      z=x-y;
```

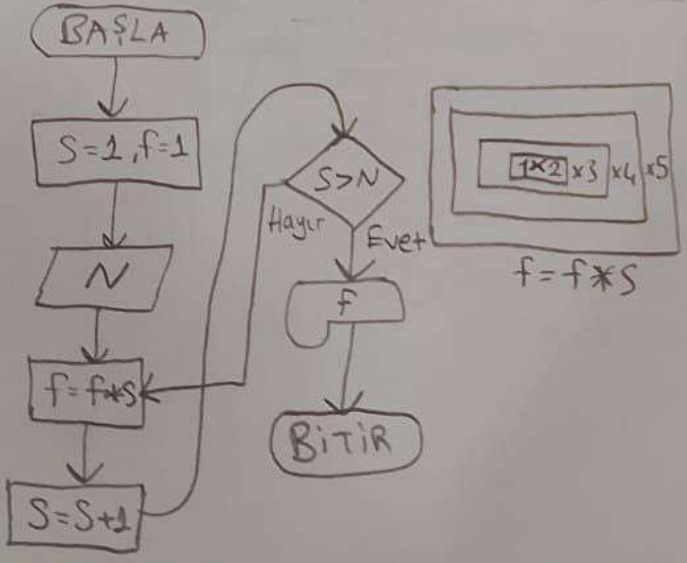


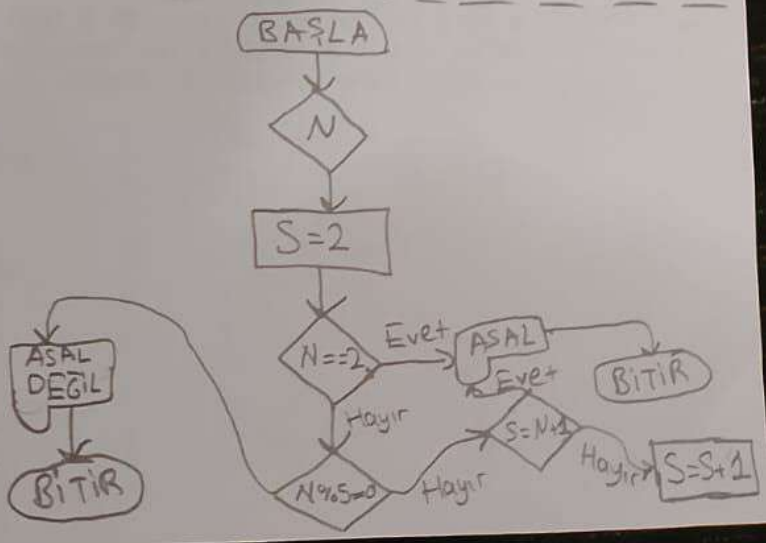


x	t
1	0
1	1
3	1
3	4
5	4









x = x+5; ⟶ x+5;
x = x/y; ⟶ $\boxed{x/=y}$ ⟶ x = x/y;
x = x%2; ⟶ x %2;
x = x+1; ⟶ x++; veya ++x;
y = y-1; ⟶ y--; veya --y;

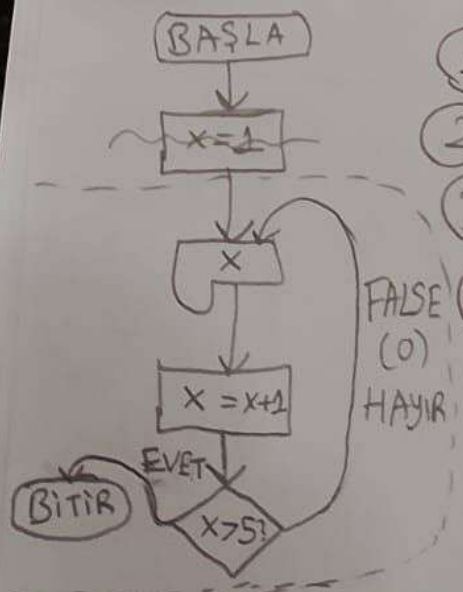
```
int x, y=4;
x= 5 + ++y;
```

```
if (x>5)
    if(y==8)
        z=1;
    else
        z=2;
else
z=3;
```

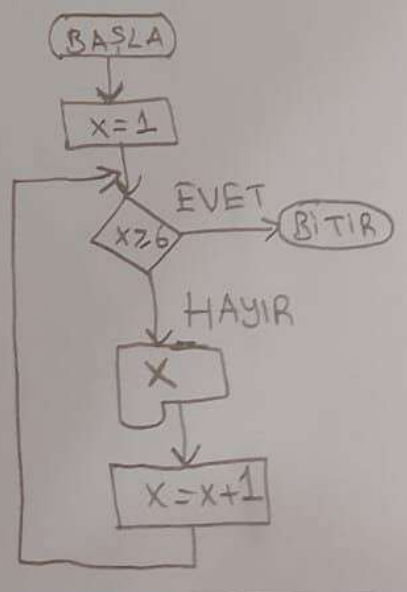
$$z = \underbrace{(x>5)}_{F}\,?\,\overbrace{(y==8)\,?\,1:2}^{T}:3$$

```
if(x>5)
    if(y==8)
        z=1;
    else
        z=2;
else
    if(y>1)
    z=3;
else if(y==1) z=4
    else z=5;
```

$$z = \underbrace{(x>5)}_{4:5}\,?\,\overbrace{(y==8)\,?\,1:2}^{T_1}:\overbrace{(y>1)\,?\,3:(y==1)?}^{F_1}$$



1. koşul
2. Döngü Kontrol Değişkeni
3. Döngü Kontrol Değişkeni'nin Güncellenmesi
4. Döngü Kontrol Değişkeni'nin ilk değeri



$(14)_{10} = (\underset{8}{1}\ \underset{4}{1}\ \underset{2}{1}\ \underset{1}{0})_2$

$14 \mid 2$ → $7 \mid 2$ → $3 \mid 2$ → $1 \mid 2$ → 0

$(14{,}25)_{10} = (\underset{2^3}{1}\ \underset{2^2}{1}\ \underset{2^1}{1}\ \underset{2^0}{0} \cdot \underset{2^{-1}}{0}\ \underset{2^{-2}}{1}\ \underset{2^{-3}}{0}\ \underset{2^{-4}}{0})$

0.5 0.25 0.120

$(0{,}875)_{10} \times 2 = 1{,}75$

$0{,}75 \times 2 = 1{,}5$

$0{,}5 \times 2 = 1{,}0$

$(0{,}111)_2$

$(0{,}6)_{10} = (?)_2$

$(0{,}1001\ 1001\ \ldots)_2$

$0{,}6 \times 2 = 1{,}2$

$0{,}2 \times 2 = 0{,}4$

$0{,}4 \times 2 = 0{,}8$

$0{,}8 \times 2 = 1{,}6$

$0{,}6 \times 2 = 1{,}2$

Enter↵

Windows

CR+LF

Carriage Return — Line Feed

'\r' '\n'

Character Buffer

B\r\n U\r\n

$A[i][j] \times B[j][k] = C[i][k]$

$i=1$ A $j=1$ — 1.B 2. — C

$$\begin{array}{c} 1. \\ 2. \\ 3. \end{array}\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix} \qquad \begin{bmatrix} 7 & 8 \\ 9 & 10 \end{bmatrix} \qquad \begin{bmatrix} \boxed{25} & \square \\ \square & \square \\ \square & \square \end{bmatrix}$$

$\overset{3}{M} \times \overset{2}{N} \qquad \overset{2}{N} \times \overset{2}{K}$

Satır Sütun

$M \times K$

i=1 j=1
i=1 j=2
i=2 j=1
i=2 j=2

```
for(int i=1; i<=3; i++) { // A'nın satırları seçer
    for(int j=1; j<=2; j++) { // B'nin sütunu seçer
        for(int k=1; k<=2; k++) {
            ([i][k] += A,[i][j] * B[j][k];
        }
    }
}
```

(0,1 001 (001 1001..)

$$Std(x) = \sqrt{\frac{1}{N} \cdot \sum_{i=1}^{N} (x_i - \bar{x})^2}$$

Ara sınav(x)

225/4
100
0
85
+ 40
56.25

$$\sqrt{\frac{1}{4} \cdot [(100-56{,}25)^2 + (0-56{,}25)^2 + (85-56{,}25)^2 + (40-56{,}25)^2}$$

İkili arama (Binary Search)

Ön kabul: Sayı dizisi sıralı olmalı!!!

n=3, aranan deger=8

dizi: 3, 6, 8, 11, 119, 22, 25, 33, 44
0 1 2 3 4 5 6 7 8

bas: ~~0~~ 2 bas
son: ~~8~~ 6
orta: ~~4~~ 3

arta (yeni) bas (yeni) son (yeni) orta Son

$C[i][j] += A[i][k] \times B[k][j]$

$$\overset{k}{\rightarrow} \quad i\downarrow \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \quad k\downarrow \begin{bmatrix} 7 & 10 \\ 8 & 11 \\ 9 & 12 \end{bmatrix} = \begin{bmatrix} \square & \square \\ \square & \square \end{bmatrix}$$

$2 \times 3 \quad 3 \times 2 \quad 2 \times 2$

$i=0 \quad j=0 \quad toplam=0 \quad k=0$

$toplam += A[0][0] \times B[0][0]$

{ "BURSA", "TEKNIK", "UNIVERSITESI", "BILGISAYAR", "MUHENDISLIGI", "BOLUMU" }

0 1 2 3 4 5

```
metin6[0] = '5';
metin6[1] = '6';
metin6[2] = '8';
metin6[3] = '10';
```

$\longrightarrow$ int x = 568;

~~int x = ato:(metin6);~~

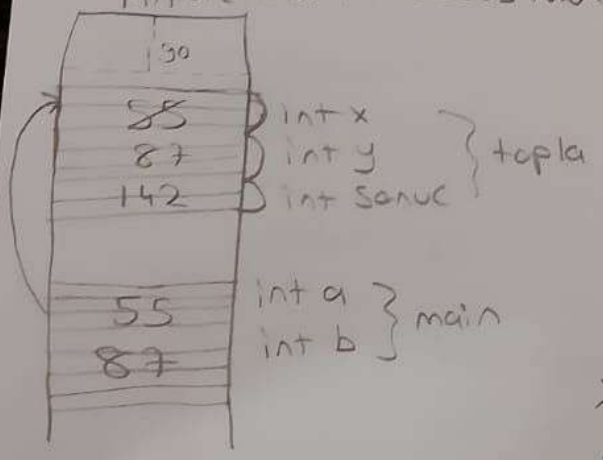
Strlen(metin6) $\longrightarrow$ 3

```
for(int i = strlen(metin6) = 1; i >= 0; i++)
    (metin6[strlen(metin6) - i - 1] - '0') * pow(10,i)
```

RAM (Random Access Memory)

pass-by-value

x = 90



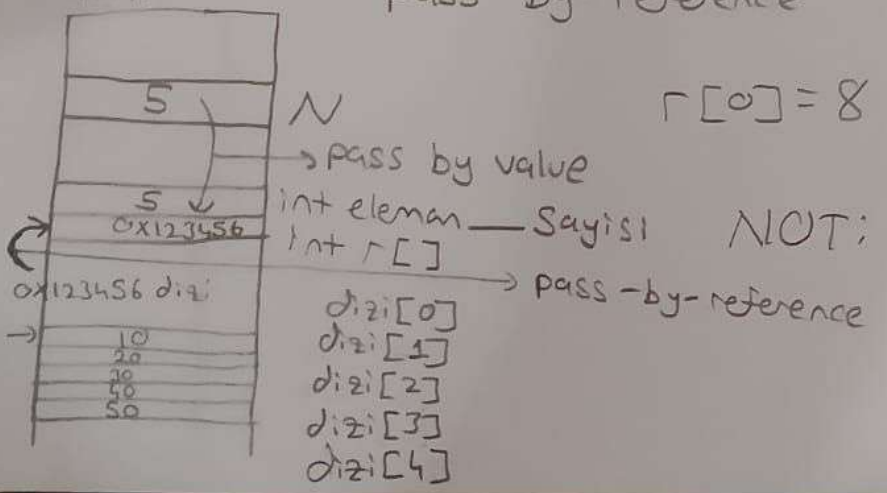
x ← a

y ← b

RAM

pass-by-reference

r[0] = 8



NOT: işin içinde dizi varsa pass by reference

int : 0 → 22 → strlen(metin1) − strlen(metin2) TEK

metin1 : "BURSA TEKNIK UNIVERSITESI"

metin2 : "TEK"

strlen(metin2) tane kontrol yapmak

a=0

---

metin 1 : "BURSA"

metin : "RS"

a = 0 1 2 BURSA (0 1 2 3 4)

b = 0 1 2

RS (0 1)

```
int faktoriyel(int N) {
N! = N * faktoriyel(N-1);
}
```

yanlış

faktoriyel(5) = 120

5! = 5 * 4! → faktoriyel(4) (N') 24

4 * 3! → faktoriyel(3) (N'') 6

3 * 2! → faktoriyel(2) (N''') 2

2 * 1! → faktoriyel(1) (N'''') → base case 1

1 * 0! → 0 * -1!

```
int faktoriyel (int N) {
    if (N==1)
        return 1;
    else
    return N* faktoriyel (N-1);
    }
```

$\frac{1}{0}, \frac{1}{1}, \frac{2}{2}, \frac{3}{3}, \frac{5}{4}, \frac{8}{5}, \frac{13}{6}, \frac{21}{7}, \frac{34}{8}, \frac{55}{9}$

$f_0 = f_1 = 1$

$f_n = f_{n-1} + f_{n-2}$

$f_9 = ?\ f_8 + f_7$

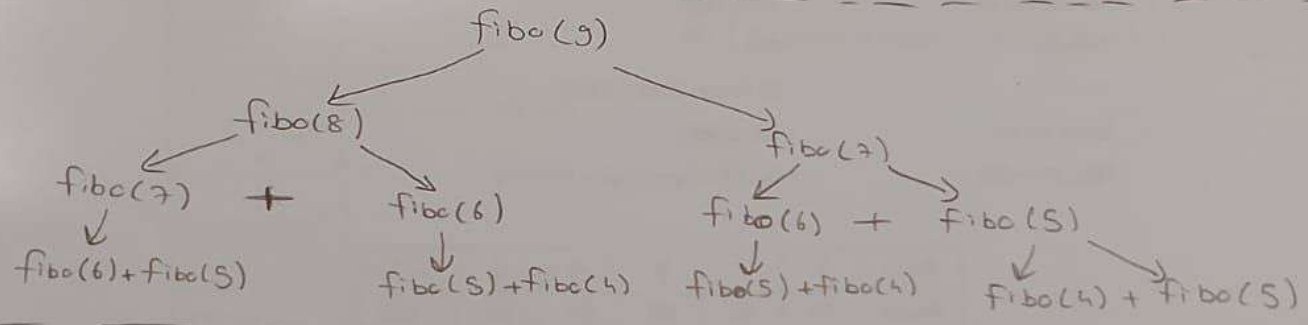
$z = x + y$

$x = y$

$y = z$

```
int x=1, y=1;
int z;
for(int i=0; i<=n; i++) {
    z = x+y;
    x=y;
    y=z;
}
printf("Serinin n indisli elemanı: %d\n", z);
```

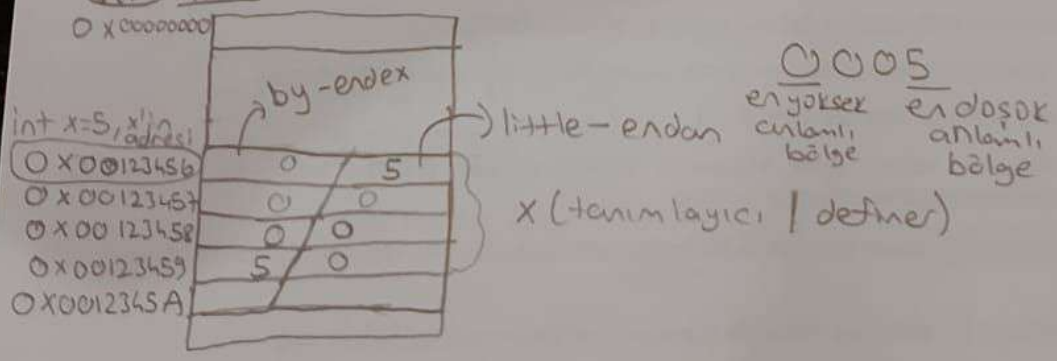
```
int fibo(int indis){
    if (indis==0 || indis==1)
        return 1;
    else
    return fibo(indis-1)-fibo(indis-2);
}
```

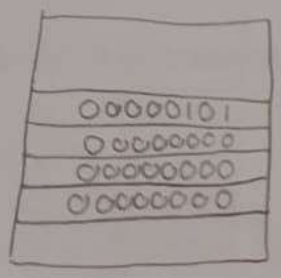


GUI: graphical user interface

CLI: Command line interface

HEV Adres





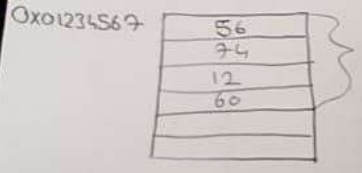
pointer (işaretçi / göstergeç): Adres tutan bir veri tipidir.

char *
int *
float *
double *

```
Chr* p = &x;
```

p = 0x0000000000061FE3C

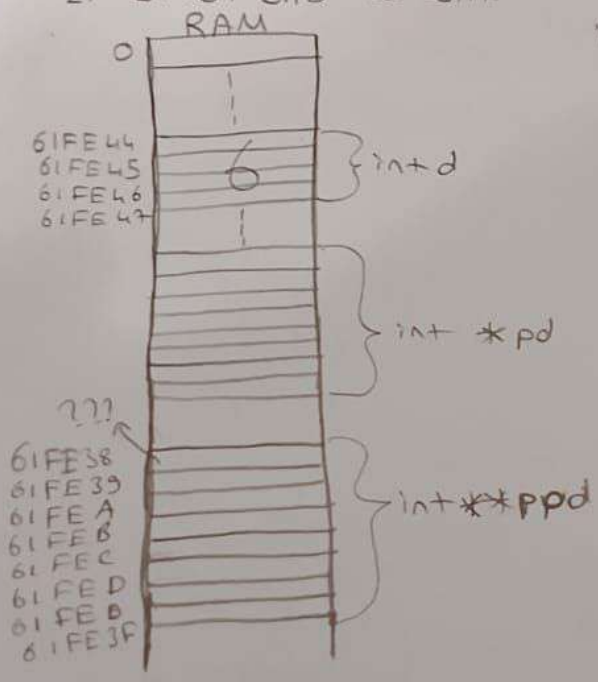
```
int* ptam = &x;
```




```
int * ptam = NULL;
```

---

dizi isimleri birer adrestir.




```
int *** pppd = &ppd;
```

ppd üzerinden 6 verilirse nasıl ulaşılır.

**ppd → **(0000000000061FE38)

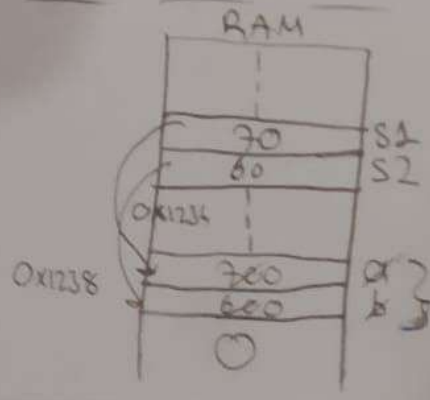
*(0000000000061FE44) → 6

---

metin1 → metin1+1

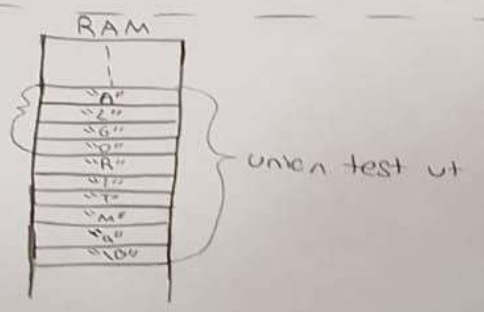
```
Char metin1[] = "BURSA";
Char metin2[100] = ;
        Strcpy(&metin2[0], &metin1[0], 3);
```

metin — metin1+1




```
int dizim[1.000.000] = { };
```

Struct hep main'in üstünde yazılır
yazarken Struct Ogrenci(gibi olanlara dikkat!!!)

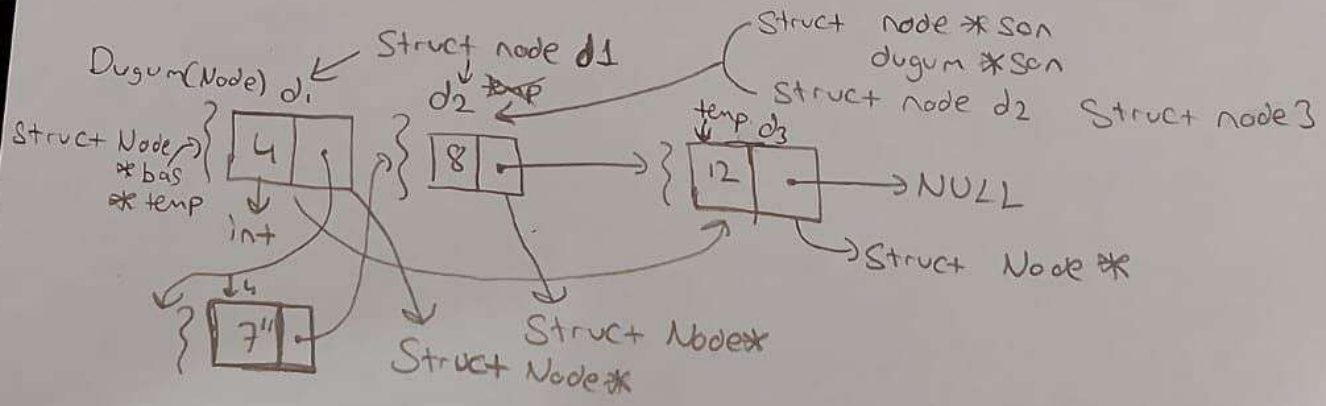


Veri Yapıları

Finalde bu çıkacak

1. Bağlı Liste (Tek | Çift Bağlı)
2. Yığın (stack)
3. Kuyruk (Queue)
4. Ağaç (Tree)
5. Yığıt (Heap)

{ 4, 8, 12, 6, 16 }
7 12
↑
7



fopen✓ fscanf✓ fgets✓ ftell✓
fclose✓ fgetc✓ rewind✓ feof✓ fread✓
fprintf✓ fputs✓ fseek✓ fwrite✓